Introduction to Symbolic Computation

MCS 320, CRN 43978, 3 credit hours

Fall 2026, MWF 1:00pm-1:50pm, SELE 2249F

About Your Instructor

Instructor Name: Jan Verschelde (You may call me Professor Jan.)
Office: SEO 1210
Office Phone: 312 996 4609
Drop-in (Office) Hours: From 2pm to 3pm on Mondays, Wednesdays, and Fridays; or by appointment. Virtual via https://uic.zoom.us/my/profjanofficehour.
Email: janv@uic.edu.
Methods of Instruction: The lectures happen in computer lab SELE 2249F.

Course Description

Introduction to Symbolic Computation via the computer algebra system SageMath. The mathematical algorithms in symbolic computation are explained with examples and applications to topics in undergraduate mathematics**.**

MCS 320 is the first course on the computational track, followed by MCS 471 (Numerical Analysis) and MCS 472 (Introduction to Industrial Math and Computation).

What You Will Learn in This Course

Course Goals and Student Learning Outcomes

The main goal of the course is to learn to apply SageMath to solve mathematical problems. The specific learning objectives are listed below.

1. Understand concepts of symbolic computation.
2. Gain hands-on experience with computer algebra.
3. Learn mathematics through computations (computational thinking).

Extensive computer use is required. This is a *computational* course, not a programming class.

Pre- or Co-requisites For This Class

The pre-requisites are MATH 210 (calculus III) and MCS 160 (introduction to computer science) or CS 107 (introduction to computing and programming); or CS 109 (programming for engineers with MatLab); or CS 111 (program design I); or consent of the instructor.

Course Website

The course web site is at https://homepages.math.uic.edu/~jan/mcs320 and a backup site is at https://janverschelde.github.io/mcs320/

On the course web site, the syllabus, lecture slides and homework assignments are posted. In this course, there will be some minimalistic use of canvas. Gradescope will be used to collect all answers to homework and exams.

Course Texts and Materials

There is no textbook for this course. Lecture notes and Jupyter notebooks will be made available electronically via the course web site.

Required Technology

We will be making extensive use of the free and open source software SageMath, available in the cloud at https://cocalc.ai. To install on your own computer, visit https://www.sagemath.org.

Answers to all homework, quizzes, projects, and exams must be submitted via gradescope in a Jupyter notebook with a SageMath kernel.

Required Reading, Video, or Lab Materials

In the summer of 2022, pre-recordings of the lectures were posted at youtube, available in the playlist https://www.youtube.com/playlist?list=PLGMCOm5grOW5rkqQ_6PymJBaKifgyWMlh

These videos are still relevant, although the current lectures may have some substantial modifications.

Recommended (Optional) Texts or Other Materials

A good reference is *Sage for Undergraduates* by Gregory V. Bard, AMS 2015 (ISBN 978-1-4704-1111-4); and available at gregorybard.com: Sage Stuff

Our Inclusive Learning Environment

UIC values diversity and inclusion. Regardless of age, disability, ethnicity, race, gender, gender identity, sexual orientation, socioeconomic status, geographic background, religion, political ideology, language, or culture, we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. If there are aspects of the instruction or design of this course that result in barriers to your inclusion, engagement, accurate assessment or achievement, please notify me as soon as possible.

Disability Accommodation Procedures

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, you should connect with the ***Disability Resource Center (DRC) at drc.uic.edu, drc@uic.edu, or at (312) 413-2183*** *to create a plan for reasonable accommodations. In order to receive accommodations, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of*

Accommodation (LOA). Letters of Accommodations can be presented at any point of the semester and, unless otherwise noted, do not expire once obtained. Upon receipt of an LOA, I will gladly work with you and the DRC to implement approved accommodations.

Community Agreement

Community Agreement: Ground Rules for a Safe/Brave Space

- *Be present (turn off cell phones and remove yourself from other distractions)*
- *Be respectful*
- *Be flexible when things don't work*
- *No eating, no snacking*
- *No side conversations*
- *Be willing to work together*
- *Be mindful of one another's privacy -- do not invite outsiders into our classroom*

This is a hand-on course, running In a computer lab, expecting you to run your own computations.

How Your Learning Will Be Assessed

There will be quizzes which will happen typically on Fridays. There will be no makeup quizzes.
If you miss a quiz or if your performance is bad, you can turn in the assigned homework to regain some of the points lost. Quizzes and homework are worth 100 points.

Exercises are assigned with each lecture. Although only a selection of the homework can be collected to make up for quizzes, it is strongly recommended to try all assignments.

Three projects will be assigned during the semester, worth jointly a total of 200 points.

During the semester, there will be two exams worth 100 points each. There will be no makeup exams given. The final exam counts for 200 points. If an exam is missed, then greater weight will be placed on the final exam, especially on the material covered on the missing exam.

Assessments

Your course grade is based on a total of 700 points: 100 from the quizzes and homework, 200 from the projects, 200 from the exams during the semester, and 200 from the final exam.

Grading scale

A	90% - 100%
B	80% - 89%
C	70% - 79%
D	60% - 69%
F	0% - 59%

Student Accountability for Coursework

When requesting an extension of a deadline to submit homework, you may expect to receive a couple of hours, not days or a week. Submitting a partial answer to homework is better than submitting nothing.

Midterm Grades

There is no formal midterm grade. By the day of the first midterm exam, after the first project, and after four weeks of classes, you should have a good understanding of first half of the materials and to a good grade in the course.

If after four weeks of classes you still feel lost in the course, then please reach out to me for help.

Final Exams

The final exam is cumulative, covering all topics in the course.

Course Policies

Tips for Success

The most important resources in this course are the Jupyter notebooks that you make with your solutions to the homework problems, the quizzes and the projects. Without sufficient practice you are likely to run out of time on the exams.

There are many ways for you to show us what you know or can do and how you are learning, through your effort, interactions, and class participation, application of critical thinking skills to solve real world problems, and performance on assignments and exams. This class has been structured to help all students get the support and guidance needed to succeed in your learning. The following tips will help you be successful in this class and in other classes throughout your career at UIC.

- *Ask questions during class*
- *Complete assignments on time*
- *Participate in class discussions*
- *Individually review your notes after class*
- *Post questions on the discussion board or ask questions during drop-in (office) hours about concepts or procedures that seemed confusing*
- *Attend / engage in drop-in (office) hours with me*

Attendance

Students are expected to attend all class meetings. Any changes in this syllabus or in the scheduling of exams and other assignments will be announced during class meetings.

Religious Holidays

I will make every effort to avoid scheduling exams or requiring student projects be submitted on religious holidays. If you wish to observe your religious holidays, please notify me by the tenth day of the semester of the date when you will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, please notify me at least five days in advance of the date when you will be absent. I will make every reasonable effort to honor your request and not penalize you for missing the class. If an examination or project is due during your absence, you will be given an exam or assignment equivalent to the one completed by those students in attendance. Students may appeal through campus grievance procedures for religious accommodations.

Class Participation / Engagement

Because active participation in the course is one of the most important ways to learn, you will be assessed on your class engagement. Engagement looks a little different for everyone, but in general, an engaged student will come to class prepared, contribute regularly to class activities or discussions, listen attentively to peers and the instructor, stay on-task during class, complete their work in a timely manner, and reach out to the instructor if they have questions or start to fall behind. If you anticipate any barriers to your full engagement in the course, I encourage you to contact me so we can strategize about how you can best fulfill the course requirements.

Your engagement or lack thereof will not be assessed.

Privacy Notification and Policy for Video Recording of Class Sessions

The recordings during the online sessions focus on the progression and narratives of the slides.

While chat discussions may contain identifying information, note that they will be available only to those enrolled in the course.

Classroom Conduct Policy

- *Be in a quiet place and use headphones*
- *Mute the microphone unless talking*
- *Video on when speaking or engaging in group activities*
- *Quit all other screens (email, text, social media) to create a classroom "presence" (also helps with internet stability)*
- *Actively participate in class*
- *Raise your hand if you have a question to ask verbally*
- *Use the chat feature if you prefer to type your question*

Policy for Late or Missing Work

Deadlines may be postponed. If you know you will be late, then it is better to apply for an extension of the deadline, instead of not submitting anything. For missed assignments, greater weight may be placed on the final exam.

Policy for Revisions

You will not be allowed to resubmit revised homework or project for a better grade.

Policy for Regrading

Any request for a regrade must be made within one week of the assignment being returned to you. If you think there has been a simple arithmetic error on your assignment then write a note explaining the error, attach this to the front of the assignment, and turn it into your TA. If the error is confirmed, then the points will be added to your score before the end of the semester. If you think there was an error in grading that is not an arithmetic error, write a brief note explaining why you think more points should be awarded, include it with the assignment, and submit it to your TA. In the event that you do turn in a regrade, you should make a copy for your own records. All regrades (except arithmetic errors) will be done at the end of the semester after final grades are calculated. If the points you request will affect your final grade, then I will reevaluate the assignment for the contested points.

Academic Integrity

No student shall claim or submit the work of another as one's own.
Allowing someone to copy from you is also punishable. All your work in this class will be individual.
Verbatim copying the output of generative AI tools is plagiarism.

UIC is an academic community committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. In this community, all members including faculty, administrators, staff, and students alike share the responsibility to uphold the highest standards of academic honesty and quality of academic work so that such a collegial and productive environment exists.

As a student and member of the UIC community, you are expected to adhere to the Community Standards of integrity, accountability, and respect in all of your academic endeavors. When accusations of academic dishonesty occur, the Office of the Dean of Students investigates and adjudicates suspected violations of this student code. Unacceptable behavior includes cheating, unauthorized collaboration, fabrication or falsification, plagiarism, multiple submissions without instructor permission, using unauthorized study aids, coercion regarding grading or evaluation of coursework, and facilitating academic misconduct. Please review the UIC Student Disciplinary Policy for additional information about the process by which instances of academic misconduct are handled towards the goal of developing responsible student behavior.

By submitting your assignments for grading you acknowledge these terms, you declare that your work is solely your own, and you promise that, unless authorized by the instructor or proctor, you have not communicated with anyone in any way during an exam or other online assessment. Let's embrace what it means to be a UIC community member and together be committed to the values of integrity.

Use of Generative Artificial Intelligence

In response to my query "use of generative ai to computer algorithm" the AI overview of Google started its reply with "Using generative AI is a highly effective way to learn computer algorithms because it acts as an interactive, 24/7 personalized tutor." giving among its references an insightful 2024 paper in the Communications of the ACM on "Computing Education in the Era of Generative AI."

Using generative AI to help you learn is good as long as you learn, that is: you gain understanding and insight in the design and analysis of algorithms. Be aware of the disclaimer: "AI can make mistakes, so double-check responses.

FERPA

Federal law (FERPA) prohibits me from disclosing or discussing your grades over e-mail. Please come see me during drop-in hours or make an appointment if you want to talk about your work.

Course Schedule

Below is the outline of the course.

Week 1: *welcome and introduction*

- *L-1 Mon 24 Aug:* welcome to MCS 320 --- SageMath and CoCalc --- computer algebra

- *L-2 Wed 26 Aug*: the Jupyter notebook --- structuring and documenting your work
- *L-3 Fri 28 Aug 28 Aug:* use as a calculator --- getting started and getting help

Week 2: *symbols and numbers*

- *L-4 Mon 31 Aug:* exact and floating-point numbers --- continued fractions
- *L-5 Wed 2 Sep:* complex and algebraic numbers --- making finite fields --- interval arithmetic
- *L-6 Fri 4 Sep:* symbols, variables, and references --- preventing evaluation

Week 3: *symbols and numbers*

- *Mon 7 Sep:* Labor Day. No classes.
- *L-7 Wed 9 Sep:* number types --- functions to store data
- *L-8 Fri 11 Sep:* evaluation and execution --- finite fields --- binary expression trees

Week 4: *vectorization, Cython*

- *L-9 Mon 14 Sep:* input/output formats --- saving and loading data from files
- *L-10 Wed 16 Sep:* timing Python functions --- vectorization --- Cython
- *L-11 Fri 18 Sep:* univariate and multivariate polynomials --- coefficient types --- monomial orders

Week 5: *polynomials and expressions*

- *L-12 Mon 21 Sep:* quotients of polynomials, expression swell, Horner form, partial fractions
- *L-13 Wed 23 Sep*: operators and operands --- expression trees for symbolic and numeric evaluation
- *L-14 Fri 25 Sep:* manipulating expressions by substitution, expansion, and factorization

Week 6: *normalization, first midterm exam*

- *L-15 Mon 28 Sep:* normalizing expressions --- canonical forms
- *L-16 Wed 30 Sep:* review to prepare for the first midterm exam
- *L-17 Fri 2 Oct:* first midterm exam

Week 7: *functions*

- *L-18 Mon 5 Oct:* defining mathematical functions --- preventing premature evaluation
- *L-19 Wed 7 Oct:* recursive functions and memoization for efficient execution
- *L-20 Fri 9 Oct:* computing with functions

Week 8: *differentiation, integration, series*

- *L-21 Mon 12 Oct:* symbolic, numeric, and implicit differentiation
- *L-22 Wed 14 Oct:* integration and summation
- *L-23 Fri 16 Oct:* series, approximations, and limits

Week 9: *plotting*

- *L-24 Mon 19 Oct:* symbolic-numeric computation
- *L-25 Wed 21 Oct:* two dimensional plots
- *L-26 Fri 23 Oct:* plotting in three dimensions and beyond

Week 10: *animations, solving equations*

- *L-27 Mon 26 Oct:* making animations
- *L-28 Wed 28 Oct:* solving equations
- *L-29 Fri 30 Oct:* linear algebra --- matrix factorizations as normal forms

Week 11:

- *L-30 Mon 2 Nov:* differential equations --- Laplace transforms
- *L-31 Wed 4 Nov:* linear programming and polyhedra
- *L-32 Fri 6 Nov:* building interactive web pages

Week 12:

- *L-33 Mon 9 Nov:* an application of interact
- *L-34 Wed 11 Nov:* review to prepare for the second midterm exam

- *L-35 Fri 13 Nov:* second midterm exam

Week 13: *sympy, numpy, and scipy*

- *L-36 Mon 16 Nov:* symbolic computing with sympy
- *L-37 Wed 18 Nov:* numeric computing with numpy and scipy

Week 14: *symbolic computing with Julia*

- *L-38 Fri 20 Nov:* introduction to Julia
- *L-39 Mon 23 Nov:* parallel computing with Julia
- *Wed 25 Nov:* Student Wellness Day. No classes.
- *Fri 27 Nov:* Thanksgiving holiday. No classes.

Week 15: *review for the final exam*

- *L-40 Mon 30 Nov:* review on the materials covered in the first midterm exam
- *L-41 Wed 2 Dec:* review on the materials covered in the second midterm exam
- *L-42 Fri 4 Dec:* cumulative review

Academic Deadlines

Important dates in the semester are listed below.

- Wednesday 23 September: Project One is due at 1pm.
- Friday 2 October: first midterm exam.
- Wednesday 28 October: Project Two is due at 1pm
- Friday 13 November: second midterm exam.
- Friday 20 November: Project Three is due at 1pm.
- Monday 7 December: final exam, from 1pm till 3pm in SELE 2249F.

Providing Feedback About This Course

At the end of the term, you will be asked to complete the course evaluations.

UIC Resources - Supporting Student Health, Wellbeing, and Academic Success

As a student at UIC, you may experience challenges such as struggles with academics, finances, student life, or your personal well-being. Please know this is completely normal and that you shouldn't hesitate to ask for help. Come to me, or if it is about an issue beyond the scope of this class, please contact your college advisors, or get help from any number of other support services and resources available to all UIC students:

- *Current Student Resources – a one-stop shop for links to resources in the following categories: General, Academic, Student Support, Student Life, Technology, Health and Safety, and Getting Around Campus*
- *UIC Tutoring Resources*
- *Programs and initiatives supporting the UIC undergraduate experience and academic programs through the Student Success Initiative*
- *If you are experiencing personal hardships (including but not limited to: personal and family emergencies, academics, interpersonal conflicts, personal safety, physical and mental health*

concerns, food or housing insecurity) and need resources and assistance, please reach out to the Student Assistance division, supported by UIC's Office of the Dean of Students.

- *Student Guide for Information Technology – a comprehensive resource for UIC students describing the most commonly used IT services and tools supporting your success*
- *Navigating Health Resources at UIC – explore the many resources that are on campus to support students' health and wellbeing and learn about how you can access them.*

Importantly, if you are in immediate distress, please know the UIC Counseling Center has crisis intervention services available 24 hours a day, 7 days a week. You can walk into the Counseling Center during business hours to request to talk to a crisis counselor, or you can call the Counseling Center main line at (312) 996-3490. If you are calling outside of business hours, select "2" during the automated greeting to be connected to a crisis counselor. You can find additional mental health resources and services on the Counseling Center Website and the Student's Guide to Accessing Behavioral Health Services at UIC.

For immediate crisis support outside of UIC, call or text "9-8-8" any time, 24 hours a day, 7 days a week. Dialing 988 will connect you to a trained crisis professional. 988 offers services in Spanish, American Sign Language, and over 240 languages through tele-interpretation services. The 988 website has an online chat feature, too.

*The **Math and Science Learning Center**, located in the Science and Engineering South Building (SES) at 845 W. Taylor St. 3rd Floor, Room 247, is a meeting place for students in Math, Biological Sciences, Chemistry, Earth and Environmental Sciences, and Physics. At the MSLC, students can meet with graduate teaching assistants for tutoring in 100-level courses, attend informal group study sessions with other students, or meet up with friends to attend one of the workshops, seminars, or other activities sponsored by the MSLC during the semester. Visit the MSLC website, call 312-355-4900, or email at mslc@uic.edu.*

*The **UIC Library** is located both on east and west campus, provides access to resources, study rooms, and research support both online via chat and in person. At Daley Library on the east side of campus, stop by the reference desk in the IDEA Commons, or make an appointment for research help on either side of campus. Learn more about library services and resources, and to find research materials in specific subject areas view the Research Guides.*

*The **Academic Center for Excellence (ACE)** can help if you feel you need more individualized instruction in reading and/or writing, study skills, time management, etc. Please call (312) 413-0031 or visit the Academic Center for Excellence (ACE) website for more information.*

***UIC Counseling Center -** Counseling services are available for UIC students who pay the student health services fee. You may seek a free and confidential intake with a trained therapist from the Counseling Center, who will work with you to create a clinically appropriate treatment plan. Treatment options may include Brief Individual Therapy or Group Therapy at the Counseling Center, referral to Psychiatry services, or Care Coordination to be connected to a community provider who meets your clinical needs. The Counseling Center is located in the Student Services Building. You may contact them at (312) 996-3490 during business hours to learn more about services, set up an intake, or for crisis intervention. Crisis intervention services are available 24/7 by walking into the Counseling Center during business hours or by calling. If you are calling after business hours, please select "2" when the automated voice message*

begins. For more information about the Counseling Center or accessing the Counseling Center and other mental and emotional health services on campus, check out the Student's Guide to Accessing Behavioral Health Services at UIC. For more information about services in the community, check out our Community Provider Database.

*The **Campus Advocacy Network** provides information and offers resources to all UIC students, faculty, and staff. Under the Title IX law, you have the right to an education that is free from any form of gender-based violence and discrimination. Crimes of sexual assault, domestic violence, sexual harassment, and stalking are against the law and can be prevented. For more information or for confidential victim services and advocacy, contact UIC's Campus Advocacy Network at 312-413-8206. To make a report to UIC's Title IX office, email TitleIX@uic.edu or call (312) 996-8670.*

***Campus Security -** As a UIC student, you've chosen to live in one of the nation's largest cities. But, as at any university, crime is a reality. At UIC, we are strongly committed to our public safety programs, and we encourage students to be proactive in learning what programs and services are available in case of an emergency. You are **discouraged** from staying in university buildings alone, including lab rooms, after hours and are **encouraged** to use the Walking Safety Escort and/or Night Ride if you are uncomfortable traveling anywhere on campus. Also, you can download the UIC SAFE app, a free personal security tool for students, faculty, and staff. It allows you easy contact with dispatchers and first responders in case of an emergency. Finally, by dialing 5-5555 from a campus phone, you can summon Police or Fire for any on-campus emergency. You may also program the complete number, (312) 355-5555, on speed dial on your phone.*

***Emergency Response Systems and Guides** - The emergency response guides can be found at the Office of Preparedness and Response. Please review and acquaint yourself with the guide and recommendations for various emergency situations.*

***Student Health Services –** If you get sick during the semester, please be aware that there are resources on campus that can support you. The Family Medicine Center is committed to providing high-quality, student centered care and services. It has attending physicians, residents, and nurse practitioners on staff to provide the full range of primary health care services. The Family Medicine Center also offers a limited set of health services at **no additional cost to students who pay the Student Health Service Fee (SHSF)**. Services include caring for an acute or urgent illness (e.g., cold, strep throat, asthma attack, flu) or an injury. More information can be found on the Family Medicine Center's Student Health webpage.*

Disclaimer

This syllabus is intended to give guidance on what may be covered during the semester and will be followed as closely as possible. However, as the instructor, I reserve the right to modify, supplement, and make changes as course needs arise. I will communicate such changes in advance as in-class announcements and in writing via email.

Last Revised: Wednesday 4 November 2026